



OMID RAHIMI

Electrical Engineer – Systems Integration, Test & RF Performance

Deggendorf, Germany • +49 176 48994467 • omid.rahimirad@gmail.com • LinkedIn: [Omidrahimi](#)

PROFESSIONAL SUMMARY

Engineer with 9+ years in systems integration, testing, and RF performance analysis across 4G/5G RAN and cloud-native 5G testbeds. KPI-driven diagnostics, reproducible test workflows, and automation using Linux/Python (pandas, scikit-learn).

Target roles: Electrical Engineer (Systems Integration/Test), Test & Validation Engineer, RF / Measurement Systems Engineer

CORE COMPETENCIES

System Integration & Validation (Systemintegration & Validierung)

Multi-component systems, interface verification, Schnittstellenprüfung, Konfigurationsvalidierung (400+ changes)

Performance Analysis (Leistungsanalyse)

Throughput, latency, reliability, KPI-driven diagnostics, SHAP-based feature analysis

Distributed & Cloud-Native Systems

5G architecture, O-RAN, Kubernetes, Docker, Linux, RF measurement & signal analysis (Mess- und Prüftechnik)

Troubleshooting & Root Cause Analysis (Fehlersuche & Ursachenanalyse)

Protocol layer analysis (RRC, NAS, MAC, PHY), OSS/NMS alarm diagnostics, Fehlerdiagnose, fault resolution

Data & Automation (Datenanalyse & Automatisierung)

Python (pandas, scikit-learn), Power BI, SQL, Git, Jupyter, automated test workflows

Languages

English (C2) · German (B2/C1) · Persian (native)

PROFESSIONAL EXPERIENCE

Research Engineer – 5G Systems & RF Technology

Deggendorf University of Applied Sciences (THD), Germany | December 2024 – March 2026

- Designed and operated a **cloud-native 5G test environment** using Kubernetes and O-RAN components for system-level performance evaluation.
- Simulated multi-user scenarios (100+ users) to analyse **system behaviour under load**, identifying bottlenecks in throughput and latency.
- Performed **end-to-end system validation**, correlating RF measurements with network KPIs to understand cross-layer effects.
- Structured automated test workflows and logs for reproducible system testing and analysis.

Wireless Systems Engineer – RAN Integration & Optimisation

Contracted via FPR Co. / Delta Ertebatat Iranian, Tehran, Iran (Huawei RAN Project) | July 2016 – March 2023

- Validated and integrated **large-scale distributed RAN systems**, ensuring correct behaviour across network components and interfaces.
- Performed **root cause analysis of system-level issues**, linking protocol behaviour to performance degradation across layers (RRC, NAS, MAC, PHY).
- Executed 50+ live system upgrades and configuration changes, ensuring **system stability and service continuity**.
- Verified 400+ configuration changes against design parameters, preventing misconfigurations in production systems.
- Analysed system performance using KPIs (throughput, latency, reliability), identifying bottlenecks and improving stability.
- Collaborated with cross-functional teams (operations, core network, vendors) to resolve complex **system integration issues**.

BSC/RNC Technical Engineer– Nationwide Network Modernisation

Iran Ofogh Industrial Development Co. / ZTE Parsian, Tehran, Iran | May 2015 – July 2016

- Integrated 900+ network nodes into existing infrastructure, ensuring system compatibility and stable operation.
- Performed configuration and parameter tuning to maintain system performance during large-scale upgrades.
- Supported infrastructure transition from legacy systems to 4G environments.

Network Operations Engineer – System Monitoring & Incident Support

SGS Iran (Huawei Technologies Service Project), Tehran, Iran | March 2013 – March 2015

- Monitored system health and performance, identifying early-stage issues through alarms and KPIs.
- Contributed to incident handling workflows and operational procedures for sustained system stability.

EDUCATION

M.Sc. Electrical Engineering & Information Technology (Automation)

Deggendorf University of Applied Sciences (THD), Germany | March 2023 – March 2026

Master’s Thesis: Permittivity Characterization for RF and Microwave Applications: Experimental Measurement and AI-Based Prediction

B.Sc. Electrical Engineering (Electronics)

Ferdowsi University of Mashhad, Iran | September 2008 – September 2012

Bachelor Thesis: Analysis of CDMA Codes: Maximal, Gold, and Kasami Sequences

PROJECTS

ML-Based Link Capacity Estimation in 4G/5G Networks

- Built predictive models (XGBoost, Random Forest) to estimate system throughput from real KPI data.
- Applied SHAP analysis to identify key system performance drivers, supporting data-driven network optimisation.

5G Test System – Cloud-Native Simulation Environment

- Deployed a complete cloud-based 5G simulation stack (core + RAN) using Kubernetes and O-RAN components.
- Automated test scenario execution and analysed stability, throughput, and latency profiles under varying loads.

CERTIFICATIONS

- Huawei Certified Network Professional (HCNP – LTE)
- Huawei Certified 5G Associate
- Python for Machine Learning (scikit-learn)
- Power BI | SQL

AWARDS & RECOGNITION

- Ranked in top 1% of Iranian National University Entrance Exam (2007)
- Technical Star Award – Huawei (2022)
- Outstanding Engineer & Network Safety Team Award (2021)
- Individual Engineer Star & Bright Star Award – Huawei (2020)

ADDITIONAL INFORMATION

- Driver's Licence: Class B | Willingness to travel: Yes
- Interests: Photography · Swimming · Fitness